$\int_3^4 \frac{2x-3}{2\sqrt{x^2-3x}(x^2-3x+e^2)}dx$

$u := x^2 - 3x \quad du = 2x - 3$

$\int_0^4 \frac{1}{2\sqrt{u}(u+e^2)}du$

$\frac{1}{2}\int_3^4 \frac{1}{u^{\frac{3}{2}}}du + \frac{1}{2e^2}\int_3^4 \frac{1}{\sqrt{u}}du$

$-\frac{1}{\sqrt{u}} \ |_3^4 + \frac{1}{e^2}\sqrt{u} \ |_3^4$

$-\frac{1}{2} + \frac{1}{\sqrt{3}} + \frac{2}{e^2} - \frac{\sqrt{3}}{e^2}$